

# ICAD RT-4 · Set 2 — Answer Sheet

Exam 12-10-2026 · 75 Q · 300 marks · 3 hours · +4 correct, -1 wrong, 0 blank

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## Physics

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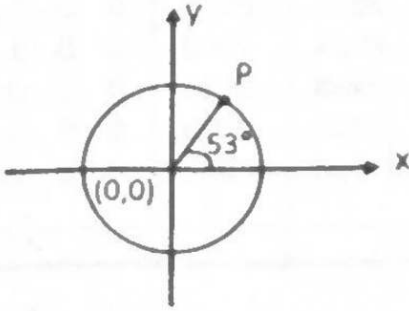
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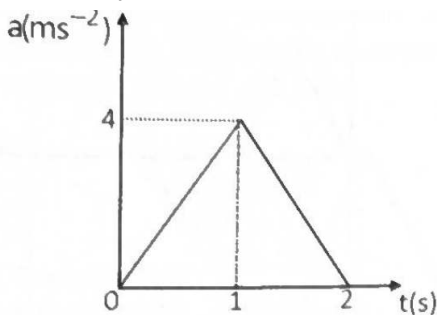
## Physics

### PART A • Q1–20 • SINGLE CORRECT

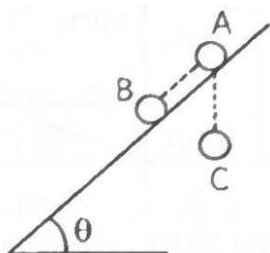
1. P1L3V4Q2 Circle of radius 10, point P at  $53^\circ$ ; coordinates of P? ((6, 8)). (a) (6, 8) (b) (8, 6) (c) (10, 10) (d) can't be determined



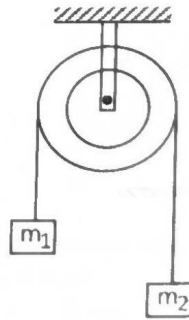
2. P2L1V4Q7 Displacement  $S = (3t^3 + 4t^2 + 10)$  m;  $ds/dt$  at  $t = 1$  s? (a) Zero (b)  $10 \text{ ms}^{-1}$  (c)  $17 \text{ ms}^{-1}$  (d)  $20 \text{ ms}^{-1}$
3. P3L1V5Q1 A point P(1, 2, 2) shifts to P'(4, 6, 3). Find the displacement vector magnitude. (a) 5 (b)  $\sqrt{26}$  (c)  $\sqrt{35}$  (d) 7
4. P3L6L7V3Q1 If area of triangle =  $1/2|A \times B|$  and  $|A \times B| = 12$ , what is the area? (a) 24 (b) 6 (c) 12 (d) 3
5. P4L2V3Q5 (Multiple correct) A car accelerates uniformly passing two checkpoints 30 m apart in 4.0 s; speed at the first checkpoint is  $5.0 \text{ ms}^{-1}$ . (a) velocity at 2nd checkpoint is  $10 \text{ ms}^{-1}$  (b) velocity at 2nd checkpoint is  $20 \text{ ms}^{-1}$  (c) acceleration is  $2.5 \text{ ms}^{-2}$  (d) acceleration is  $1.25 \text{ ms}^{-2}$
6. P4L6V3Q5 Acceleration-time graph of a particle (triangle 0-2s, peak 4 at  $t=1$ ). Velocity at  $t=0$  is  $2 \text{ ms}^{-1}$ . The velocity after 2 s will be (a) 6 (b) 4 (c) 2 (d)  $8 \text{ ms}^{-1}$



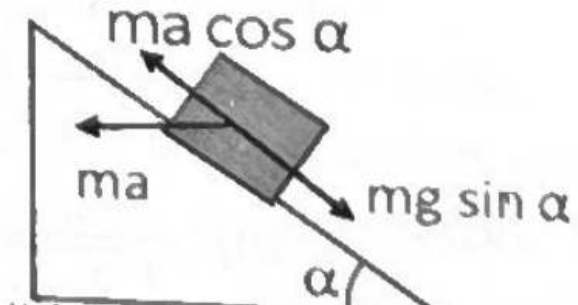
7. P4L10V3Q3 A 100 m train at  $15 \text{ ms}^{-1}$  overtakes a man running on the platform in the same direction in 10 s. How long would it take to cross him if he ran in the opposite direction instead? (a) 7 s (b) 5 s (c) 3 s (d) 1 s
8. P4L12V5Q9 A particle is released on a smooth inclined plane from point A, and another particle is simultaneously dropped from the same point to fall freely. If B and C are the positions of the two particles after time  $t$ , then: (a)  $AB \perp AC$  (b)  $AB \perp BC$  (c)  $AC \perp BC$  (d) None of these



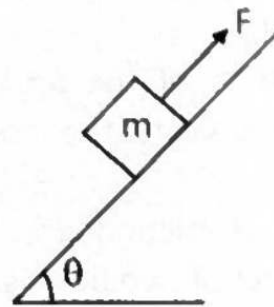
9. P5L1V5SuggQ5 In a pulley-mass system with frictionless pulley & massless string, if  $m_1 = m_2$ , the thrust on the pulley will be: (a) less than  $(m_1 + m_2)g$  (b) greater than  $(m_1 + m_2)g$  (c) equal to  $(m_1 + m_2)g$  (d) none of these



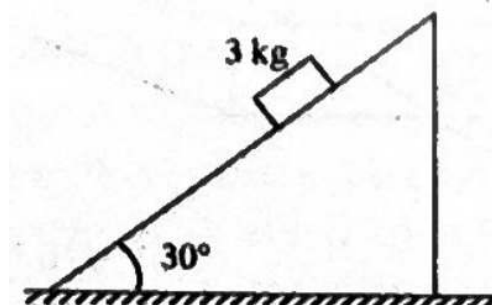
10. P6L7V3Q5 Block on a frictionless incline  $\alpha$ ; incline accelerated by  $a$  to keep it stationary —  $a = (g / g \tan \alpha / g / \tan \alpha / g \csc \alpha)$



11. P7L1V3Q3 A force  $F$  is applied on a wooden block placed on a rough incline, directed up along the incline as shown. The direction of friction on the wooden block: (a) must be up the incline (b) must be down the incline (c) may be up the incline (d) none of these.

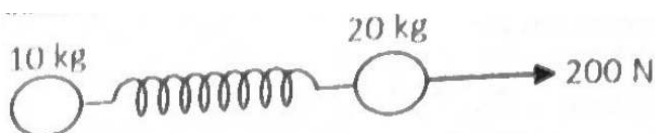


12. P7L1V5Q1 A block of mass 3 kg is at rest on a rough fixed inclined plane of angle  $30^\circ$ . The magnitude of the net force exerted by the surface on the block will be ( $g = 10 \text{ m/s}^2$ ): (a) 26 N (b)  $15\sqrt{3}$  N (c) 15 N (d) 30 N.



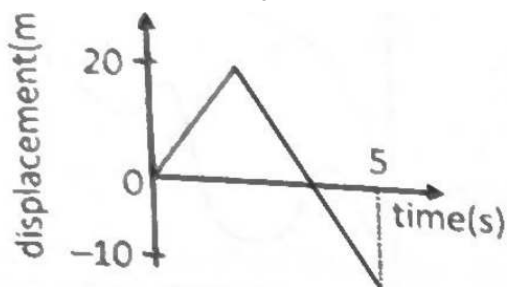
13. P8L1V5Q7 A block of mass 1 kg moves from point A(2 m, 3 m, 4 m) to point B(3 m, -4 m, 8 m). During the motion of the block, a force  $F = (2)\hat{i} + (4)\hat{j} - (10)\hat{k}$  N acts on it. Find the work done by the force  $F$  on the block. (a) 66 J (b) -66 J (c) -90 J (d) 90 J

14. P8L2V4Q9 A force  $\mathbf{F} = (3t\mathbf{i} + 5\mathbf{j})$  N acts on a body due to which its displacement varies as  $\mathbf{s} = (2t^2\mathbf{i} - 5\mathbf{j})$ . Work done by this force in  $t = 0$  to  $2$  sec is: (a) 23 J (b) 32 J (c) zero (d) can't be obtained
15. P8L5V3Q5 At two positions the kinetic energy and potential energy of a particle are  $K_1 = 10$  J,  $U_1 = -20$  J,  $K_2 = 20$  J,  $U_2 = -10$  J. In moving from 1 to 2: (A) work done by conservative forces is positive (B) work done by conservative forces is negative (C) work done by all the forces is positive (D) work done by all the forces is negative. Options: (a) Only (A) and (B) (b) (A), (B) and (C) (c) Only (A) (d) O
16. P6L5V5Q1 10 kg & 20 kg connected by a massless spring; 200 N acts on the 20 kg mass; at that instant the 10 kg mass has a  $a = 12$  m/s<sup>2</sup>. Find the acceleration of the 20 kg mass.  
(write the value)

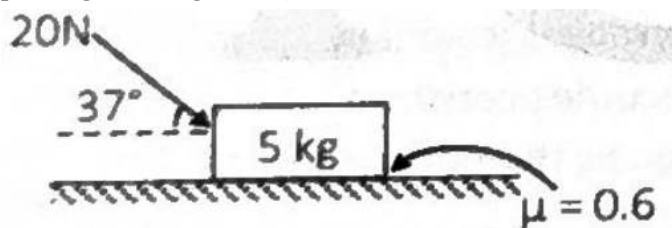


## PART B · Q21–25 · NUMERICAL VALUE

21. P3L5V3Q5 Given  $\mathbf{A} = \hat{i} - 2\hat{j} - 3\hat{k}$  and  $\mathbf{B} = 4\hat{i} - 2\hat{j} + 6\hat{k}$ , calculate the angle made by  $(\mathbf{A} + \mathbf{B})$  with the x-axis in degrees.
22. P4L6V2Q1 Displacement-time graph for a particle in a straight line (rises to 20 m then falls to -10 m). Find average velocity (in m/s) for the interval  $t = 0$  to  $t = 5$

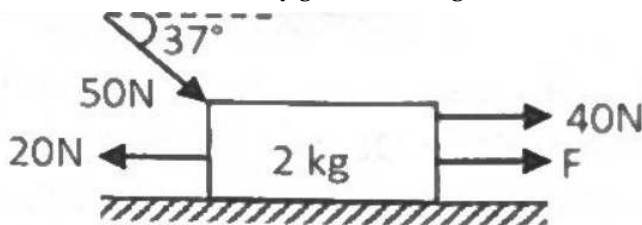


17. P7L2V3Q2 Find the friction force ( $F_r$ ) acting on the block and its direction if any, in newtons. A 5 kg block on a rough floor ( $\mu = 0.6$ ) is pushed by a 20 N force inclined at  $37^\circ$  to the horizontal (horizontal component leftwards, vertical component pressing down). ( $g = 10$  m/s<sup>2</sup>) (write the value)



18. P8L7V3Q1 A block of mass  $m$  moving at a speed  $v$  compresses a spring through a distance  $x$  before its speed is halved. The spring constant of the spring is given by  $(a \cdot m \cdot v^2) / (b \cdot x^2)$ ; then find 'ab'. (write the value)
19. P6L7V3Q1 Apparent weight  $w_1$  (up) and  $w_2$  (down, same  $a$ ); weight at constant speed  $= (w_1 + w_2) / x$ . Find  $x$ . (write the value)
20. P8L1V3Q2 A gardener pulls a lawn roller along the ground through a distance of 20 m. If he applies a force of 20 kg wt in a direction inclined at  $60^\circ$  to the ground, find the work done by him (in J). ( $g = 9.8$  m/s<sup>2</sup>) (write the value)

23. P3L6L7V6Q10 A particle moves so its position vector  $\mathbf{r}$  has constant magnitude  $r$ , but its direction rotates with angular rate  $\Omega$  about a fixed axis  $\hat{n}$ . Which option(s) are correct? (e.g. areal rate  $= |\mathbf{r} \times (d\mathbf{r}/dt)| / 2 = r^2 \Omega / 2$ )
24. P4L12V6Q10 (Multiple Correct) Wind blows towards East at  $30$  ms<sup>-1</sup>. Airspeed of an aeroplane is  $50$  ms<sup>-1</sup>. The plane must go from city A to city B, 1200 km due North. Choose the correct statement(s) about the heading direction ( $37^\circ$  West of North) and flight time (500 min / 100 min).
25. P6L2V3Q2 Normal reaction by ground on 2 kg block.



## Chemistry

### PART A · Q1–20 · SINGLE CORRECT

1. C2L3V3Q3 When intensity increases, what happens to the max KE of photoelectrons? (a) Increases (b) Decreases (c) Becomes zero (d) Remains same
2. C2L3V5Q1  $K_{\max} = 2.40 \times 10^{-20}$  J,  $\nu_0 = 1.10 \times 10^{15}$  Hz — calculate incident frequency  $\nu$ . (a)  $1.10 \times 10^{15}$  Hz (b)  $1.14 \times 10^{15}$  Hz (c)  $7.47 \times 10^{14}$  Hz (d)  $3.62 \times 10^{13}$  Hz
3. C2L6V5Q3 Difference in angular momentum of two successive Bohr orbits. (a)  $h$  (b)  $h/2\pi$  (c)  $2h$  (d)  $2\pi h$
4. C2L8V4Q6 Cricket ball 0.5 kg at  $v = 100$  m/s — associated wavelength. (a) 1/100 cm (b)  $6.6 \times 10^{-34}$  m (c)  $1.32 \times 10^{-35}$  m (d)  $6.6 \times 10^{-28}$  m
5. C3L2V3Q1 Calculate the number of oxygen atoms in 3 mol of H<sub>2</sub>O. (a)  $3 \times 6.022 \times 10^{23}$  (b)  $6.022 \times 10^{23}$  (c)  $1.8066 \times 10^{24}$  (d)  $9.033 \times 10^{23}$
6. C3L12V4Q6 Different proportions of oxygen in the various oxides of nitrogen, prove the law of: (a) reciprocal proportions (b) multiple proportions (c) constant proportions (d) conservation of mass
7. C3L12V4Q7 3 g of a hydrocarbon on combustion in excess of oxygen produces 8.8 g of CO<sub>2</sub> and 5.4 g of H<sub>2</sub>O. The data illustrates the law of: (a) conservation of mass (b) multiple proportions (c) constant proportions (d) reciprocal proportions
8. C4L1V5SuggQ1 According to Lavoisier's classification, elements that have a tendency to gain electrons were placed in which category? (a) Metals (b) Non-metals (c) Metalloids (d) Transition elements
9. C4L2V5SuggQ4 Between F and Ne, which element has the greater effective nuclear charge ( $Z^*$ ) for valence electrons? (a) Ne, because  $\sigma$  is smaller (b) F, because  $Z$  is smaller (c) Ne, because  $Z$  increases but  $\sigma$  increases only slightly (d) F, because shielding is maximum in F
10. C4L3V3Q2  $K^+$ ,  $Ca^{2+}$ ,  $Cl^-$  are isoelectronic. Arrange in increasing radius. (a)  $Ca^{2+} < K^+ < Cl^-$  (b)  $Cl^- < Ca^{2+} < K^+$  (c)  $K^+ < Ca^{2+} < Cl^-$  (d)  $Cl^- < K^+ < Ca^{2+}$

11. C5L1V3Q2 Which change will decrease lattice energy when anion remains constant? (a) Increase in cation charge (b) Decrease in cation size (c) Increase in cation size (d) Increase in electronegativity difference
12. C5L4V4Q4 Which hybridisation has sulphur in  $\text{SO}_2$ ? (a)  $\text{sp}^2$  (b)  $\text{sp}^3\text{d}^2$  (c)  $\text{sp}^3$  (d)  $\text{sp}$
13. C5L6V5Q8 Decreasing order of bond angle of (I)  $\text{NH}_3$ , (II)  $\text{PH}_3$  and (III)  $\text{AsF}_3$  is: (a) I > II > III (b) I > III > II (c) II > I > III (d) III > II > I
14. C5L9V3Q5 Dipole moment is most useful in distinguishing between which of the following? (a) Atomic and molecular compounds (b) Ionic and covalent solids (c) Cis and trans isomers of the same compound (d) Sigma and pi bonds
15. C5L9V4Q3 The molecule having nonzero dipole moment is: (a)  $\text{CH}_4$  (b)  $\text{BF}_3$  (c)  $\text{NF}_3$  (d)  $\text{CO}_2$
16. C2L7V4Q10 Max emission lines when excited electron at  $n=5$  drops to ground state. (a) 12 (b) 15 (c) 21 (d) 10
17. C3L6V4Q8  $2\text{Al} + \text{Fe}_2\text{O}_3 \rightarrow 2\text{Fe} + \text{Al}_2\text{O}_3$ : 3 mol Al + 1 mol  $\text{Fe}_2\text{O}_3$ . Limiting reagent and max moles Fe. (a) Al, 2 mol (b) Al, 1 mol (c)  $\text{Fe}_2\text{O}_3$ , 1 mol (d)  $\text{Fe}_2\text{O}_3$ , 2 mol
18. C4L2V4Q7 For Ca ( $Z = 20$ ), configuration  $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$ , calculate the shielding constant ( $\sigma$ ) for a 4s electron. (a) 18.85 (b) 17.15 (c) 19.70 (d) 21.35
19. C5L2V3Q1 Calculate the total number of valence electrons in a CO (carbon monoxide) molecule ( $C = 4$ ,  $O = 6$ ). If a triple bond shares 6 electrons, how many electrons are left to be used as lone pairs? (a) 4 electrons (b) 3 electrons (c) 6 electrons (d) 2 electrons
20. C1L7V4Q8 Four gases, equal V & P, diff densities — which is  $1/2$  u He & which greatest u rms? (a)  $\text{O}_2$  has half the speed; He greatest (b)  $\text{N}_2$  has half the speed; He greatest (c)  $\text{CO}_2$  has half the speed; He greatest (d) None has half; He greatest

## PART B · Q21–25 · NUMERICAL VALUE

21. C1L7V3Q5 RMS speed of a gas,  $M=2.5$ , at  $50^\circ\text{C}$ . (a) 1500 m/s (b) 1700 m/s (c) 1800 m/s (d) 2000 m/s
22. C3L2V4Q9 1 electron has charge  $1.6022 \times 10^{-19}$  C; calculate the charge on 0.1 mole  $\text{Zn}^{2+}$  ions
23. C1L2V3Q1 Fahrenheit equivalent of  $100^\circ\text{C}$ ? (a) 212 F (b) 100 F (c) 150 F (d) 180 F
24. C1L3V4Q1 3.2 g gas = same volume as 0.2 g  $\text{H}_2$  ( $17^\circ\text{C}$ ); gas at  $27^\circ\text{C}$  — molecular weight. (a) 22.4 (b) 28.0 (c) 33.1 (d) 44.8
25. C1L5V2Q3 Density  $A=9$ ,  $B=4$  — ratio of diffusion rates  $r_A/r_B$ . (a) 2:3 (b) 3:2 (c) 1:1.5 (d) 1.5:1

## Mathematics

### PART A · Q1–20 · SINGLE CORRECT

1. M1L1V6Q3  $57^\circ 17' 45''$  in radians  $\approx$ ? (a) 0.95 (b) 1.00 (c) 1.05 (d) 1.10
2. M2L2V2Q4 600 students: 400 Telugu, 300 Hindi  $\rightarrow$  how many speak both. (a) 100 (b) 200 (c) 300 (d) 400
3. M4L1V4Q3  $A = \{x: 1 \leq x < 10\}$ ,  $B = \{x: 0 \leq x \leq 12\} \rightarrow A \cup B, A \cap B$ . (a)  $A \cup B = (0, 12]$  (b)  $A \cap B = [1, 10]$  (c)  $A \cap B = (1, 10]$  (d) None of these
4. M4L2V4Q6 In  $X = \{a, b, c, d\}$ , which of  $R_1 \dots R_4$  are functions? (a)  $R_1 = \{(b, a), (a, b), (c, d), (a, c)\}$  (b)  $R_2 = \{(a, d), (d, c), (b, b), (c, c)\}$  (c)  $R_3 = \{(a, b), (b, c), (c, d), (b, d)\}$  (d)  $R_4 = \{(a, a), (b, b), (c, c), (a, d)\}$
5. M5CL1V4Q8  $(\cos A + \sin A)/(\cos A - \sin A) - (\cos A - \sin A)/(\cos A + \sin A) = ?$  (a)  $2 \tan 2A$  (b)  $2 \operatorname{cosec} 2A$  (c)  $2 \cot 2A$  (d)  $2 \sec 2A$
6. M5AL3V2Q5 Evaluate the maximum value of  $f(\theta) = 8 \sin \theta + 15 \cos \theta$ . (a) 15 (b) 17 (c) 23 (d) 10
7. M5CL4V4Q7  $A + B + C = 360^\circ \Rightarrow \tan(A/2) + \tan(B/2) + \tan(C/2) =$  (a)  $\sin(A/2) \sin(B/2) \sin(C/2)$  (b)  $\cos(A/2) \cos(B/2) \cos(C/2)$  (c)  $\tan(A/2) \tan(B/2) \tan(C/2)$  (d)  $\cot(A/2) \cot(B/2) \cot(C/2)$
8. M6L1V5SuggQ4 For which interval of  $x$  does the inequality  $(x-1)/(x+2) < 0$  hold? (a)  $(-2, 1)$  (b)  $(-\infty, -2)$  (c)  $(1, \infty)$  (d)  $(-\infty, -2) \cup (1, \infty)$
9. M6L3V4Q4 If  $|x - 2| \leq |x + 4|$ , then: (a)  $x \in [0, \infty)$  (b)  $x \in (-\infty, 1)$  (c)  $x \in [1, \infty)$  (d)  $x \in [-1, \infty)$
10. M6L3V5Q10 For  $a > 0$ , consider  $|x - 3a| + |x - a| \leq 2|x - 2a|$ . Which statement is true? (a) Solution set is  $(-\infty, a] \cup [3a, \infty)$  (b) Solution set is  $[a, 3a]$  (c) Solution set is  $\mathbb{R}$  (d) No real solution
11. M7L3V4Q1 If  $\log_4 5 = a$  and  $\log_5 6 = b$ , then  $\log_3 2$  is equal to: (a)  $1/(2a+1)$  (b)  $1/(2b+1)$  (c)  $2ab+1$  (d)  $1/(2ab-1)$
12. M7L4V4Q8 If  $5^{(3x^2 - \log_{10} 2)} = 2^{((x+1/2) \cdot \log_{10} 25)}$ , then  $x$  equals: (a) 1,  $-1/3$  (b) 1 (c) 1,  $-1/2$  (d)  $1/3, -1$
13. M8AL2V4Q3 If the ratio of sum of  $n$  terms in two A.P.'s is  $2n : n + 1$ , then the ratio of 8th terms is (a) 15 : 8 (b) 8 : 13 (c) 5 : 17 (d) none
14. M8BL1V3Q3 If 2nd term of a G.P. is 24 and 5th term is 81 then the first term is (a) 16 (b) 18 (c) 21 (d) 12
15. M8BL3V1Q2 If  $a, ar, ar^2$  are in G.P., then  $1/a, 1/ar, 1/ar^2$  are: (a) Also in G.P. (b) In A.P. (c) Constant (d) Random
16. M5CL2V3Q5  $(\cos \alpha - \cos \beta)^2 + (\sin \alpha - \sin \beta)^2 = k \sin^2((\alpha - \beta)/2) \Rightarrow k =$  (a) 4 (b) 3 (c)  $3/2$  (d)  $1/4$
17. M7L1V3Q2 If  $2^{(x-7)} \times 5^{(x-4)} = 1250$ , then the value of  $x$  is: (a) 2 (b) 4 (c) 8 (d) 16
18. M8BL2V3Q1 How many terms of the Series  $1 + 2 + 2^2 + \dots$  must be taken to make sum 511. (write the value)
19. M8AL3V5Q1 A man saves Rs. 32 during the first year, Rs. 36 in the second year and in this way he increases his savings by Rs. 4 every year. Find in what time (in years) his saving will be Rs. 200. (write the value)
20. M8CL1V4Q1 If  $a, b, c$  are in H.P. and  $ab + bc + ac = 21$ , then the value of 'ac' is (write the value)

### PART B · Q21–25 · NUMERICAL VALUE

21. M6L3V4Q8 If  $|4 - x| + 1 < 3$ , then  $x \in (a, b)$ ; find  $|b/a|$ .
22. M8AL2V3Q1 If the sum of  $1 + 4 + 7 + 10 + \dots + x = 590$ , then  $x$  is
23. M3L2V4Q1  $[\cos(\pi+x)\cos(-x)]/[\sin(\pi-x)\cos(\pi/2+x)] = k \rightarrow 2k \cdot \tan^2 x$
24. M5BL1V4Q7 If  $\cos 40^\circ + \cos 80^\circ + \cos 160^\circ + \cos 240^\circ = k$ , then  $3k =$  \_\_\_\_\_
25. M8AL2V3Q2 If 4 terms are added between 2 and 12 so that the result sequence form an A.P. then average of these term is ...